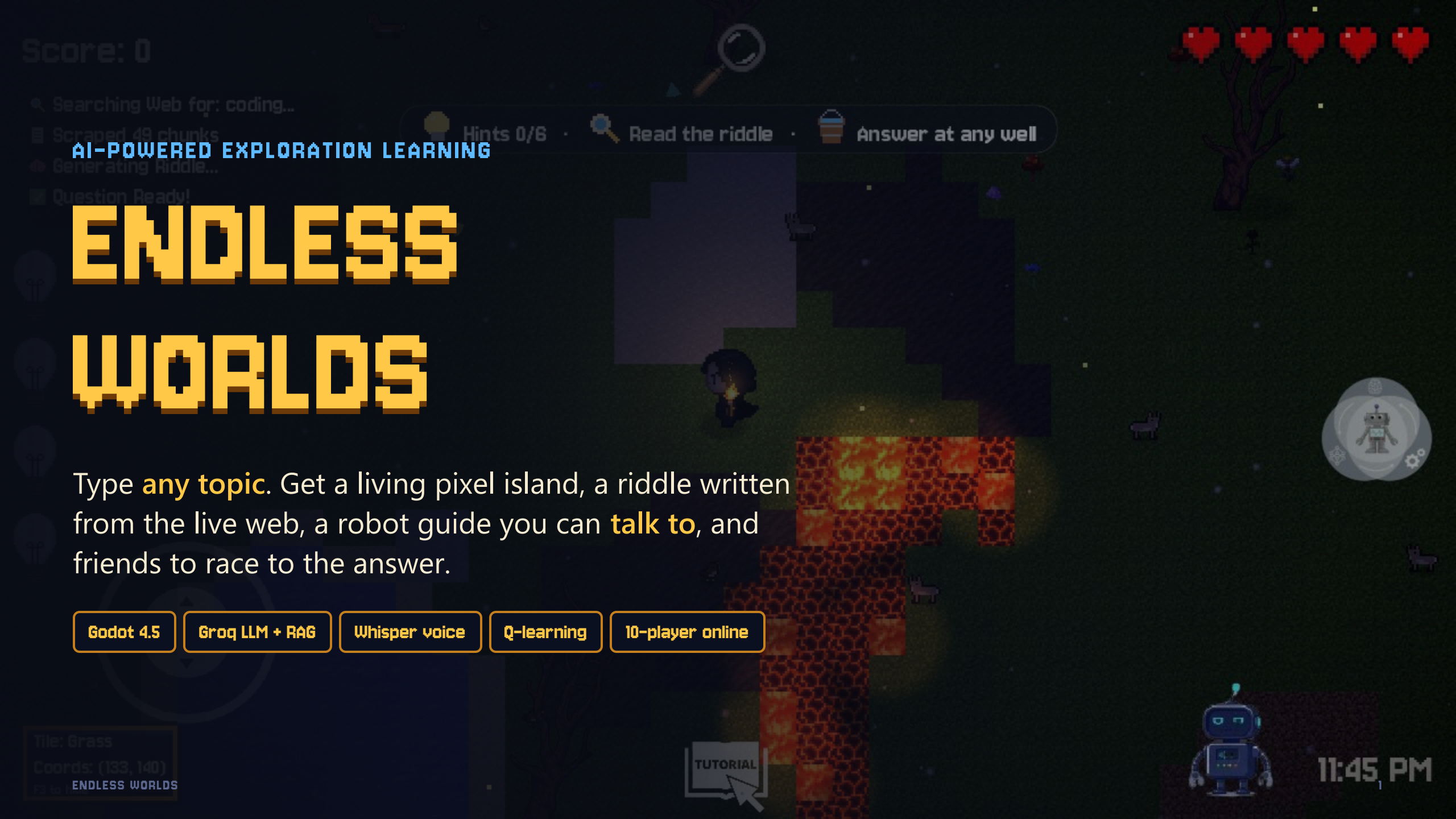

Score: 0
Searching Web for: coding...
Scraped 49 chunks
Generating Riddle...
Question Ready!
Hints 0/6
Read the riddle
Answer at any well
AI-POWERED EXPLORATION LEARNING
ENDLESS WORLDS
Type any topic. Get a living pixel island, a riddle written from the live web, a robot guide you can talk to, and friends to race to the answer.
Godot 4.5
Groq LLM + RAG
Whisper voice
Q-learning
10-player online
Tile: Grass
Coords: (133, 140)
ENDLESS WORLDS
TUTORIAL
11:45 PM

THE PROBLEM

Most learning games are quizzes in costume

Fixed content

A finished question bank. Once you've seen it, the "game" is over, and it can't teach a topic nobody wrote questions for.

One-size difficulty

The same level for everyone: too easy gets boring, too hard makes players quit. Nothing adapts.

Learning alone

No voice, no companion and no friends, just a screen of multiple choice.

What the research says about AI companions

Studies of AI companions in serious games report more engagement when the AI **scaffolds with hints and questions instead of giving answers**, and warn that learners can over-rely on an AI that solves things for them.[1] Endless Worlds follows that rule: its bot is prompted never to say the answer.

[1] e.g. SPARC/CAP multiplayer serious-game pilot (Frontiers in Computer Science); AI verbal & visual scaffolding in a serious game (arXiv 2602.08893); 90-participant human-agent collaboration study (arXiv 2605.27384).

One loop: explore → collect → answer → level up

TOPIC
"black holes", "Python", "Rome"… or tap a chip

→

NEW ISLAND
procedural terrain, 4 biomes, weather, day/night, wildlife

→

RIDDLE
written from a live web page by the LLM, at your difficulty

→

EXPLORE
compass → 6 hint bulbs; avoid lava, water; ask Bolt out loud

→

ANSWER
at any well, in one of 6 mini-game formats

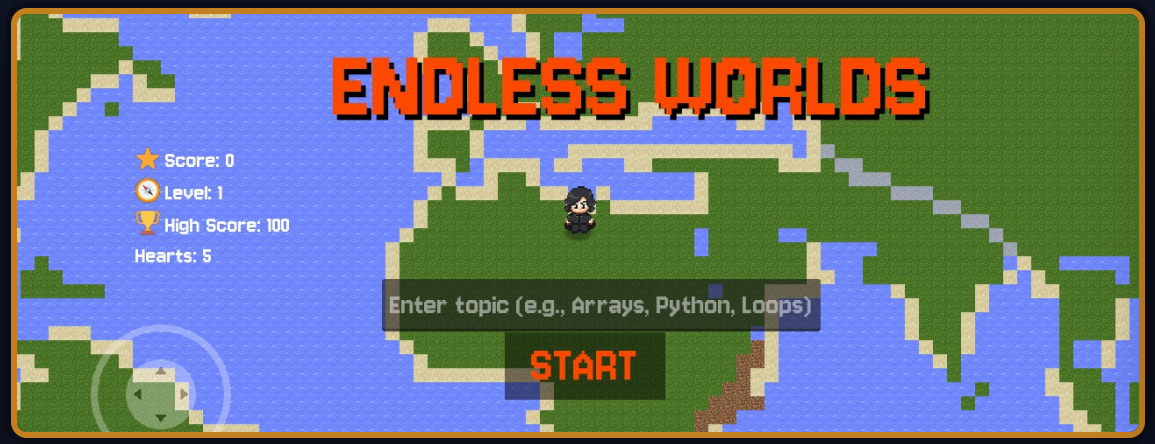
win → fact saved to Learning Journal → streak bonus → harder next level



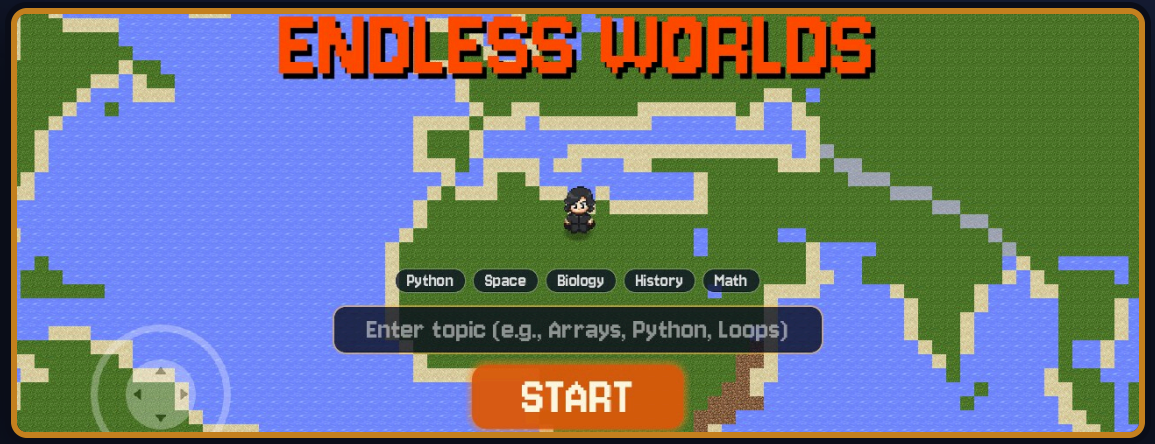


BEFORE → AFTER

Same game, levelled up



Before: flat menu, plain input



After: topic chips, live AI status + RL difficulty, multiplayer, pixel icons



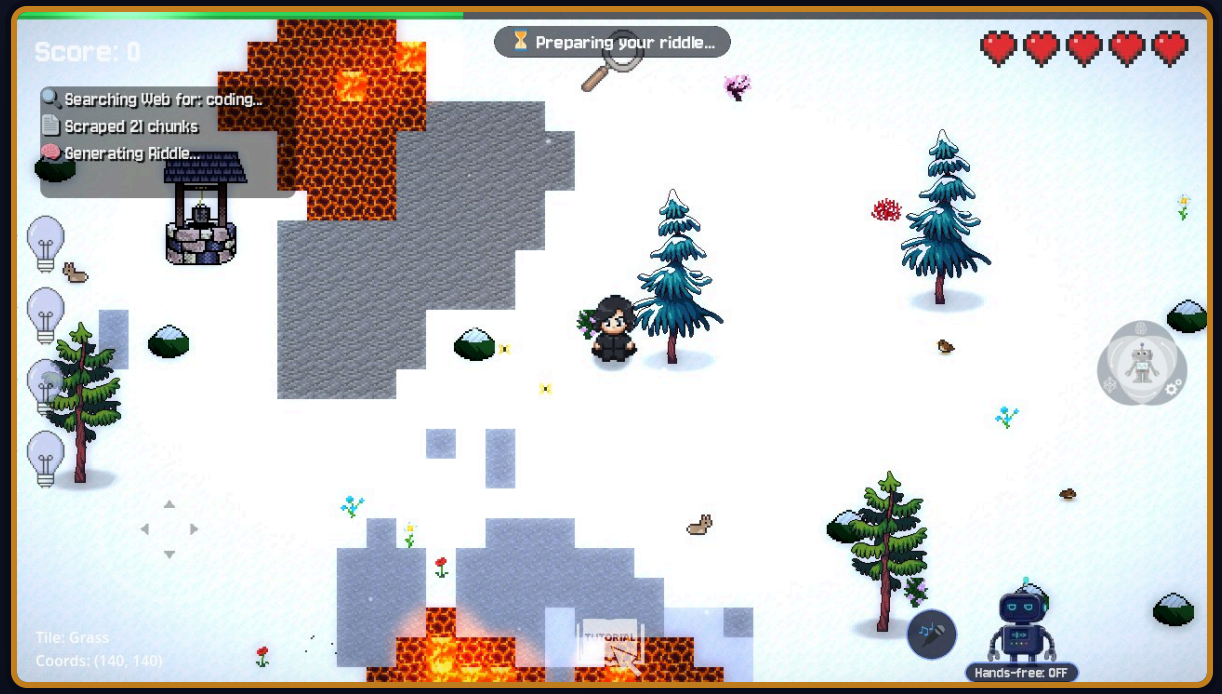
Before: no goal shown, stuck on "Searching Web..." if the server was off



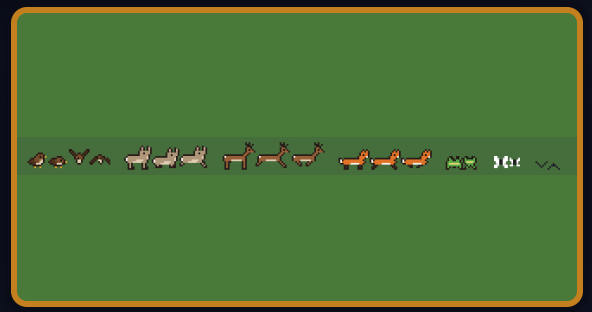
After: objective bar, compass to hints/wells, torch lighting, living wildlife

A WORLD THAT'S ALIVE

Every island reacts to you







- **Wildlife:** sparrows peck and fly off when anyone gets close; rabbits hop away and turn at the shoreline; butterflies by day, fireflies by night, flies buzzing
- **More life:** deer and foxes bolt, frogs hop along the shore, and V-shaped flocks of birds cross the sky with soft shadows on the ground
- **Jump:** Space for an arc with squash-and-stretch and landing dust; it clears small lava patches and syncs online
- **Real snow:** round drifting flakes instead of slowed-down rain streaks
- **Lighting:** CanvasModulate night with real point lights: a hand torch (auto at night, F to toggle), glowing wells and hint bulbs, and lava light
- **Post-processing:** bloom on lava and flames, vignette, colour grade per biome, film grain
- **9 characters** incl. Dhoti, Chudidar and Saree outfits, re-pixelized to the world's pixel grid, picked on the home screen and synced online
- All sprites are **original pixel art** drawn for this game, so there are no licence issues

Ask out loud. Get a nudge, not the answer.

MIC
hold V · or hands-free voice-activity detection with pre-roll

→

WAV 16 kHz
AudioEffectCapture downsampled in GDScript

→

WHISPER
/api/transcribe
Groq whisper-large-v3-turbo · 0.47 s

→

RAG GUIDE
/api/rag + riddle
"never say the answer"
≈ 1 s reply

→

SPEAKS
bubble + OS/browser text-to-speech;
reads hints aloud



Designed to stay out of the way

- Click Bolt and a **pixel-icon tray** slides out: talk, hands-free, type, voice on/off, torch, chat
- A pulsing red ring follows your mic level while he listens
- The speech bubble grows upward and never covers the controls
- Offline? He still gives a helpful tip instead of an error
- Real test: a spoken "what is a tuple in Python?" was transcribed word-for-word and answered without naming the riddle's answer

THE CONTENT ENGINE

Any topic → a riddle grounded in a real web page

SEARCH
DuckDuckGo → Bing → Brave fallback chain

SCRAPE
5 pages in parallel, first clean one wins (nav/ads stripped)

CHUNK+EMBED
500-char chunks, MiniLM-L6 vectors (384-d)

FAISS
one index per topic, LRU cache of 64 → repeat topics instant

LLM
Groq gpt-oss-120b, JSON mode, topic + RL difficulty

VALIDATE
1-word answer ≤12, answer ∈ options, 4 hints · 1 retry

GAME
riddle, options, hints, fact → journal

OFFLINE PATH

Server down or slow (25–30 s timeouts)? The game switches instantly to a topic-matched built-in bank of 38 validated riddles. It never hangs.

≈2 s	0 s	4/4
riddle generation (measured)	scrape for a cached topic	validator unit tests pass

ADAPTIVE DIFFICULTY

A Q-learning agent keeps you in the flow zone



Five levels from **VERY_EASY** to **VERY_HARD**. After every game the agent updates its Q-table:

```
Q ← Q + 0.3 · (r + 0.5 · max Q′ − Q)
```

- The chosen difficulty is sent to the LLM, so the riddle itself gets harder or easier
- Persisted to `user://difficulty_rl.json` across sessions
- Fixed in this version: three separate RL copies were writing the same save file, and wrong answers were fed back twice. Now there is **one model and one update per game**.

Streak bonus +10 / win **Learning Journal**

VARIETY

Six answer formats + 50 coding quests

Multiple choice

Classic four options from the LLM.

Fill in the blank

Type it: recall beats recognition.

Wordle

Coloured letter feedback; each wrong row costs a heart.

Whack-a-answer

Hit the right word as options pop out of holes.

Word lock

Spin letter wheels to spell the answer.

Quiz show

Gold-and-blue stage, 50:50, audience poll and phone-a-friend lifelines, dramatic lock-in.

Coding mode

50 programming quests checked by an LLM "compiler" that explains what's right or wrong. They're picked at random and use the same hearts, score and stats.

PLAY TOGETHER

Up to 10 players, one island, live



Live over the Cloudflare relay: coloured name tags, shared hint, chat bubble, scoreboard

- **Room codes**: create, share 5 letters, join. Works across cities with no port forwarding
- **One master laptop**: the host's game picks the world seed and generates the riddle; everyone builds the identical island from **one number**
- Shared hints, chat (T), first correct answer wins the round, auto next round
- Everyone keeps the **character they picked**, from Shadow to Saree
- Guests talk to Bolt **through the host's AI**, so they don't need Python
- Relay: Cloudflare Worker + Durable Object, **free plan**, deployed and tested

Full technical walkthrough: **Multiplayer-How-It-Works.pdf**

ARCHITECTURE

Small, separate parts

GODOT 4.5 CLIENT (GDScript)

World
simplex terrain · biomes · wells
wildlife · flocks · weather
time system · atmosphere/lights

Play
player · hearts · tasks/hints
6 answer popups · coding
guide HUD · round card

AI glue
question generator + offline bank
DifficultyRL (Q-learning)
Bolt + VoiceAssistant

Online
Net autoload (WebSocketPeer)
MultiplayerSync · puppets
Lobby · host voice proxy

Global (autoload)
save/load · stats · journal · streak · API/relay URLs from .env · pixel icon font
Exports: Windows .exe (one file) · Web (HTML5, single-threaded for GitHub Pages)

HTTP

WSS

PYTHON AI SERVER (FastAPI)

/api/scrape · /api/generate_riddle · /api/rag
/api/transcribe · /api/evaluate_code · /health

ddgs search · BeautifulSoup · sentence-transformers
FAISS per topic · Groq LLM + Whisper · validator

sync endpoints (thread pool) · CORS · size limits
Dockerfile → Hugging Face Spaces / Render

CLOUDFLARE RELAY (JS)

Worker + Durable Object per room · hibernation
10-player cap · sender stamping · 768 KB msgs
114 lines · integration-tested locally + live

ENGINEERING QUALITY

Bugs found and fixed in this pass

Problem	Effect before	Fix
Three DifficultyRL copies sharing one save file	Feedback double-counted; copies overwrote each other	One model in Global, one update per game
Win recorded before +30 score / level up	Stats and best level always one win behind	Single `_record_victory()` in the right order
No HTTP timeouts; no offline path	Stuck on "Searching Web…" forever if the server was down	Timeouts + instant offline bank
One global vector index on the server	Two players' topics overwrote each other	Per-topic FAISS + LRU cache
Blocking work inside `async` endpoints	One slow scrape froze every request	Sync endpoints in the thread pool
LLM output trusted blindly	Multi-word answers broke Wordle; answer missing from options	Validator + repair + 1 retry, unit-tested
Footsteps loaded via `DirAccess`	Silent footsteps in exported builds	`ResourceLoader.list_directory`
Player could spawn in lava; spawn pad used the wrong biome tiles	Instant damage; mismatched tile patch	Hazard-free spawn circle in the biome's tileset
Hard-coded `localhost:8000`, no CORS	Couldn't be hosted; web build couldn't reach the API	URL from .env / project / page, CORS

SHIPPING

One script, three deliverables, free hosting

Players

`EndlessWorlds-Windows.zip`: one 108 MB .exe with the relay address built in. Unzip and play solo or online.

Host's AI

`EndlessWorlds-AI-Server.zip`: double-click `start_server.bat`. Or run the Dockerfile on Hugging Face Spaces for everyone.

Browser

`build/web`: single-threaded HTML5 export, so it runs on GitHub Pages or itch.io without special headers.

tools/package.ps1

`powershell -ExecutionPolicy Bypass -File tools\package.ps1 -Godot <godot.exe>`: imports, exports Windows + Web, zips the game and the AI server. Verified: two packaged .exe copies joined a room over the live relay and shared a riddle and hints.

$0

Cloudflare Workers Free relay

$0

GitHub Pages / itch.io web build

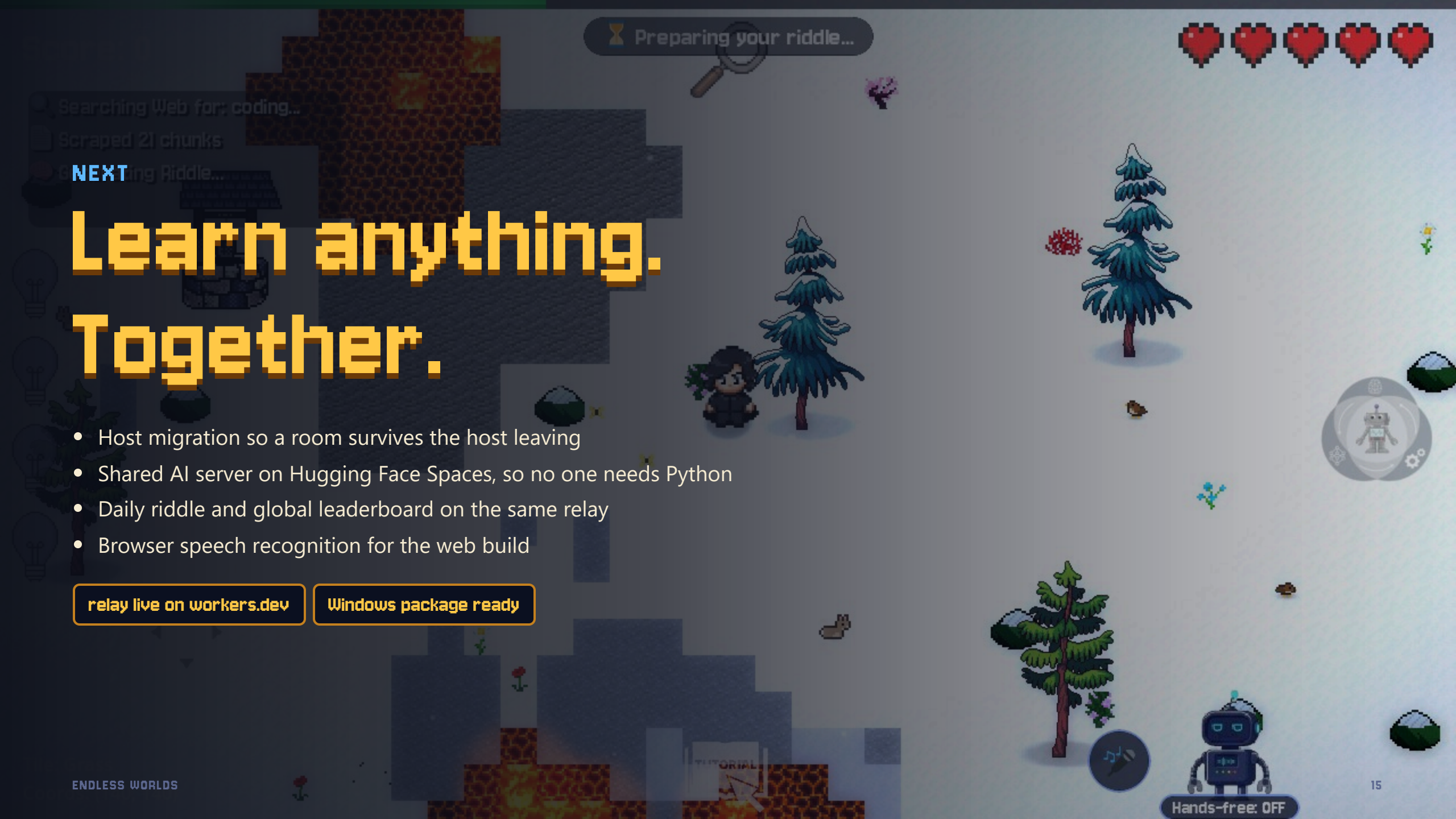
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Groq free tier · HF Spaces CPU

WHY IT'S BETTER

Endless Worlds vs. the usual

	Typical quiz app	Typical edu game	Endless Worlds
Topics	fixed bank	a few hand-made levels	any topic, sourced from the live web
Difficulty	static	manual levels	Q-learning adapts per player
Help	show answer	static hint text	voice companion that nudges, never spoils
World	none	fixed map	new procedural island every round, living wildlife
Formats	1	1–2	6 mini-games + 50 coding quests
Social	leaderboard	rarely	10-player live rooms with chat
Offline	often fails	yes	yes: built-in bank, no hangs
Memory	score only	progress bar	Learning Journal of every solved riddle + fact



Learn anything. Together.

- Host migration so a room survives the host leaving
- Shared AI server on Hugging Face Spaces, so no one needs Python
- Daily riddle and global leaderboard on the same relay
- Browser speech recognition for the web build

relay live on workers.dev | Windows package ready